

Written By: Tony Cruise

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Introduction

What I aim to do in this series of articles is to provide an insight into machine code programming on your computer. The articles will take for granted that you know how to program in BASIC, but will not require knowledge of machine code.

Full machine code listings will be included for the person who wants to know more, but they will not be necessary for understanding the article.

The series of programs will eventually add extra commands that you can use in your BASIC programs.

The sections will include:-

- Making use of all 64k of your computer's memory

- Printer dump routines

- Screen save and load to memory and/or tape and disk

- Automatic sprite movement with velocities
- Sprite collision detection
- Screen scrolling routines

Part 1 – Using More Memory

A lot of the original MSX game titles that used more than 32K of RAM were very machine dependant. A classic example is “Finders Keepers” which only works on the Sony HIT-BIT.

The author took it for granted that all of the MSX machines would have their RAM in the same area.

To explain this further look at the diagram below:

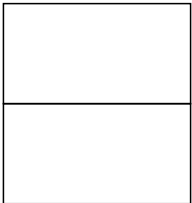
0000H

Slot 0

Slot 1

Slot 2

Slot 3



ROM



SONY

RAM

--

Page 0

4000H

ROM

--

SONY

RAM



Page 1

8000H



ROM



SONY

RAM



Page 2

C000H

ROM



SONY
RAM



Page 3

FFFFH



Any of the boxes in the picture can hold 16K of RAM or ROM (Up to 64K can be placed in each box,

but that is beyond the scope of this article).

Only one of the four slots can be activated for each page of memory.

Thus when you turn on the computer the BASIC ROM is activated, so only the top 32K of RAM can be accessed.

This is why you only get 28K of RAM free for your BASIC programs (The other 4K of RAM is used by BASIC as a storage

space for system variables e.g. cursor position).

Machine Code Program

	1	ORIGIN	08400H	
	2	:		
	3	:	PROGRAM TO FIND EXTRA	
	4	:		
F3	5	STABIT		; Disables all interrup
DB	A8	IN A,(0A8)		; Get page settings
4F	7	LD C,A;		Store in C
06	03	LD B,3;		Loop counter
21	0000	LD HL,0;		Memory location to
C5	10	LP	PUSH B;	Save BC
79	11	LD A,C;		
B0	12	OR B	:	

CB20	SLA B ;
CB20	SLA B ;
B0 15	OR B ; Calculate slot
D3AB	OUT (0A8H),A; bank informa
C1 17	POP BC; Restore BC
3E08	LD A,1 ;
77 19	LD (HL),A
7E 20	LD A,(HL)
3D 21	DEC A ;
28022	JR Z,EXIT; it RAM
10E23	DJNZ L; Do loop again
78 24 EXIT	LD A,B ;
B1 25	OR C ;
CB26	SLA B ;
CB20	SLA B ;
B0 28	OR B ; Calculate slot

32	29	FF	LD (0FF, B70H), A	0FFE7H
79	30		LD A,C; Set pages to normal	
D3	A8		OUT (0:A8H), A	
FB	32		EI ; Enable interrupts	
C(	33		RET ; Return to BASIC	

Basic Loader

```
10 CLS: CLEAR
```

```
200, &HCFFF: DEFINT A-Z: A  
=&HD000
```

```
20 READ A$: IF A$ <> "@"
```

```
THEN POKE A,
```

```
VAL("&H"+A$): A=A+1: GOT
```

```
O 20
```

```
30 PRINT" INSERT
```

TAPE/DISK TO SAVE
PROGRAM”

40 PRINT” AND
PRESS ANY KEY”

50

A\$=INPUT\$(1):PRINT:PRI
NT “ SAVING”

60

BSAVE”CAS:RAMSEL”,&H

D000,A-1:REM Remove

CAS: if using disk

100 DATA

F3,DB,A8,4F,06,03,21,00,0

0,C5

110 DATA

79,B0,CB,20,CB,20,B0,D3,
A8,C1

120 DATA

3E,01,77,7E,3D,28,02,10,E
C,78

130 DATA

B1,CB,20,CB,20,B0,32,E7,
FF,79

140 DATA D3,A8,FB,C9

150 DATA @